

53-amaliy mashg'ulot

1-misol. Ushbu

$$f(x, y) = \sqrt{1 - x^2 - y^2}$$

funksiyaning $(0,0)$ nuqtada qat'iy maksimumga erishishi ko'rsatilsin.

◀ $\delta > 0$ ($0 < \delta < 1$) sonni olib, $(0,0)$ nuqtaning $U_\delta((0,0))$ atrofini hosil qilamiz.

Unda $\forall (x, y) \in U_\delta((0,0)) \setminus \{(0,0)\}$ uchun

$$f(x, y) = \sqrt{1 - x^2 - y^2} < f(0,0) = 1$$

bo'ladi. Demak, berilgan funksiya $(0,0)$ nuqtada maksimumga erishadi. ▶

2-misol. Ushbu

$$f(x, y) = x^2 + xy + y^2 - 2x - 3y$$

funksiya ekstremumga tekshirilsin.

◀ Avvalo berilgan funksiyaning statsionar nuqtalarini topamiz:

$$f'_x(x, y) = 2x + y - 2, \quad 2x + y - 2 = 0, \quad x_0 = \frac{1}{3}$$

$$f'_y(x, y) = x + 2y - 3, \quad x + 2y - 3 = 0, \quad y_0 = \frac{4}{3}.$$

Demak, $\left(\frac{1}{3}, \frac{4}{3}\right)$ statsionar nuqta.

Ravshanki,

$$f''_{x^2}(x, y) = 2, \quad f''_{xy}(x, y) = 1, \quad f''_{y^2}(x, y) = 2.$$

Demak,

$$a_{11} = 2, \quad a_{12} = 1, \quad a_{22} = 2$$

$a_{11} = 2 > 0$ va $a_{11}a_{22} - a_{12}^2 = 3 > 0$ bo'lganligi uchun berilgan funksiya $\left(\frac{1}{3}, \frac{4}{3}\right)$

nuqtada lokal minimumga erishadi va

$$\min f(x, y) = f\left(\frac{1}{3}, \frac{4}{3}\right) = -\frac{7}{3}$$

bo'ladi. ►

3-misol. Ushbu

$$f_1(x, y) = x^4 + y^4,$$

$$f_2(x, y) = -(x^4 + y^4),$$

$$f_3(x, y) = x^3 + y^3,$$

funksiyalar ekstremumga tekshirilsin.

◄ Berilgan funksiyalar uchun $(0,0)$ statsionar nuqta bo'ladi. Bu funksiyalar uchun

$$a_{11}a_{22} - a_{12}^2 = 0$$

bo'ladi. Ravshanki, $(0,0)$ nuqtada $f_1(x, y)$ funksiya minimumga, $f_2(x, y)$ funksiya esa maksimumga erishadi. $f_3(x, y)$ funksiya $(0,0)$ nuqtada ekstremumga ega bo'lmaydi. ►

Mashqlar

1. $f(x, y)$ funksiya uchun (x_0, y_0) statsionar nuqtada

$$f''_{x^2}(x_0, y_0) \cdot f''_{y^2}(x_0, y_0) - [f''_{xy}(x_0, y_0)]^2 < 0$$

bo'lsa, $f(x, y)$ funksiya (x_0, y_0) nuqtada ekstremumga erishmasligi isbotlansin.

2. Ushbu

$$f(x, y) = (y - x)^2 + (y + 2)^3$$

funksiya ekstremumga tekshirilsin.

Quyidagi ko'p o'zgaruvchili funksiyalarni ekstremumga tekshirilsin:

1. $z = x^2 + (y - 1)^2$

2. $z = (x - y + 1)^2$

3. $z = x^2 y^3 (6 - x - y)$

4. $z = x^4 + y^4 - x^2 - 2xy - y^2$

5. $z = xy \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} \quad (a > 0, b > 0)$

6. $z = 1 - \sqrt{x^2 + y^2}$

$$7. z = e^{x^2-y} (5 - 2x + y)$$

$$8. z = \sin x \sin y \sin(x + y)$$

$$9. z = x^2 + xy + y^2 - 4 \ln x - 10 \ln y$$

$$10. z = xy \ln(x^2 + y^2)$$

11. $u = (x + y + z)e^{-(x+2y+3z)}$ funksiyaning $x > 0, y > 0$ sohada inf va sup topilsin.